

Amino Acid Chemistry

Definition, classification, D/L forms, and functions

Core idea:

Amino acids are organic compounds containing amino and carboxyl functional groups. In proteins, the standard building blocks are mostly L-alpha-amino acids. The side chain or R group gives each amino acid its unique chemical nature.

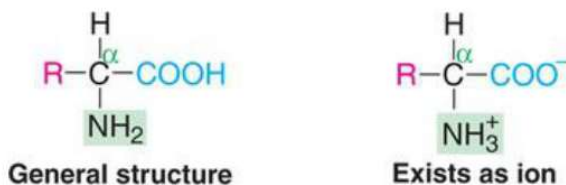
1. Definition of Amino Acids

Amino acids are a group of organic compounds containing two functional groups - an amino group (-NH₂), which is basic, and a carboxyl group (-COOH), which is acidic. In biological fluids, the amino group commonly exists as -NH₃⁺ and the carboxyl group as -COO⁻.

Proteins on complete hydrolysis yield L-alpha-amino acids; therefore, proteins are polymers of L-alpha-amino acids. Of the many amino acids present in nature, 20 standard amino acids are commonly incorporated into proteins by the genetic code.

2. General Structure of Amino Acids

An alpha-amino acid has the amino group, carboxyl group, hydrogen atom and a variable side chain or R group attached to the same alpha-carbon atom. The R group determines the identity, charge, polarity and function of the amino acid.



General alpha-amino acid structure and ionized form from the provided source.

Part	Name	Importance
-NH ₂ / -NH ₃ ⁺	Amino group	Basic group; accepts H ⁺ and gives amino acids amphoteric nature.
-COOH / -COO ⁻	Carboxyl group	Acidic group; donates H ⁺ and forms peptide bonds.
H	Hydrogen atom	One of the four groups attached to the alpha-carbon.
R	Side chain	Variable group responsible for each amino acid's properties.

3. D and L Amino Acids

Except glycine, amino acids have an asymmetric alpha-carbon atom and therefore show optical isomerism. D and L forms are assigned by comparison with D- and L-glyceraldehyde. Protein amino acids in humans are L-alpha-amino acids.

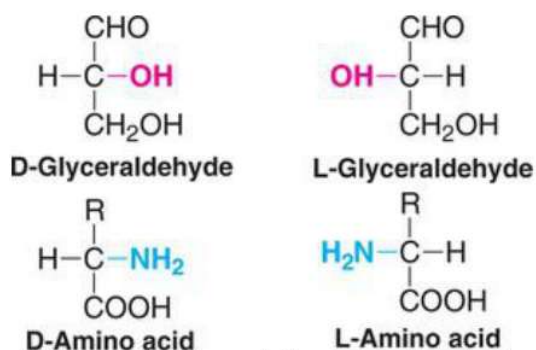


FIG. 4.1 D- and L-forms of amino acid based on the structure of glyceraldehyde.

D- and L-forms of amino acid based on glyceraldehyde from the provided source.

Form	Important points
L-amino acids	Major form in proteins of humans, animals and plants; used in normal protein synthesis.
D-amino acids	Rare in higher organisms; found in some antibiotics, bacterial cell walls, and small amounts in brain tissue.
Glycine exception	Glycine has R = H, so its alpha-carbon is not asymmetric; it has no D/L optical isomerism.

4. Classification of Amino Acids with Examples

A. Classification based on structure and chemical nature

Class	Examples	Exam point
Aliphatic side chain	Glycine, Alanine, Valine, Leucine, Isoleucine	Valine, leucine and isoleucine are branched-chain amino acids.
Hydroxyl group containing	Serine, Threonine, Tyrosine	Can participate in hydrogen bonding; tyrosine is also aromatic.
Sulfur containing	Cysteine, Methionine	Cysteine has -SH; two cysteines can form cystine/disulfide bond.
Acidic amino acids and amides	Aspartic acid, Glutamic acid; Asparagine, Glutamine	Acidic amino acids carry extra carboxyl group; amides are their derivatives.
Basic amino acids	Lysine, Arginine, Histidine	Have basic side chains; positively charged at physiological pH.
Aromatic amino acids	Phenylalanine, Tyrosine, Tryptophan	Contain aromatic rings; absorb ultraviolet light.
Imino acid	Proline	Has pyrrolidine ring and imino group; creates bends in protein chains.
Heterocyclic amino acids	Histidine, Tryptophan, Proline	Contain ring structures with atoms other than carbon.

B. Classification based on polarity

Group	Examples	Importance
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Group	Examples	Importance
Non-polar / hydrophobic	Alanine, Leucine, Isoleucine, Valine, Methionine, Phenylalanine, Tryptophan, Proline	Usually buried inside globular proteins.
Polar uncharged	Glycine, Serine, Threonine, Cysteine, Glutamine, Asparagine, Tyrosine	Participate in hydrogen bonding.
Polar positively charged	Lysine, Arginine, Histidine	Basic side chains; bind negatively charged molecules.
Polar negatively charged	Aspartate, Glutamate	Acidic side chains; important in enzyme active sites and salt bridges.

C. Nutritional classification

Class	Examples	Meaning
Essential / indispensable	Valine, Leucine, Isoleucine, Lysine, Methionine, Phenylalanine, Threonine, Tryptophan	Cannot be synthesized sufficiently; must be supplied in diet.
Semi-essential	Arginine, Histidine	Needed especially during growth; adults may synthesize some amount.
Non-essential / dispensable	Glycine, Alanine, Serine, Cysteine, Aspartate, Asparagine, Glutamate, Glutamine, Tyrosine, Proline	Synthesized by the body according to need.

Memory tip:

Essential amino acids can be remembered as PVT TIM HALL: Phenylalanine, Valine, Tryptophan, Threonine, Isoleucine, Methionine, Histidine, Arginine, Leucine, Lysine. In many exam answers, Histidine and Arginine are noted as semi-essential.

D. Classification based on metabolic fate

Class	Examples	Fate
Glycogenic	Alanine, Aspartate, Glycine, Methionine and many others	Carbon skeleton can form glucose or glycogen.
Ketogenic	Leucine, Lysine	Carbon skeleton forms acetyl-CoA/acetoacetate and can contribute to ketone bodies or fat.
Both glycogenic and ketogenic	Isoleucine, Phenylalanine, Tryptophan, Tyrosine	Can contribute to glucose as well as ketone/fat synthesis.

5. Zwitterion, Ampholyte and Isoelectric pH

Amino acids contain both acidic and basic groups, so they can act as ampholytes: they can donate or accept protons depending on pH. In physiological solution, amino acids usually exist as zwitterions or dipolar ions, having both positive and negative charges but no net charge at the isoelectric pH.

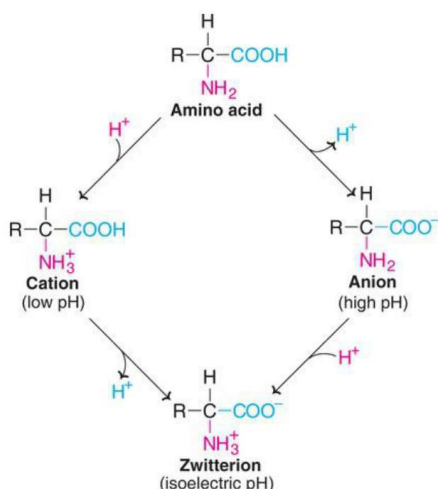


FIG. 4.2 Existence of an amino acid as cation, anion and zwitterion.

Amino acid as cation, anion and zwitterion from the provided source.

Condition	Ionic form
Low pH / acidic medium	Amino acid accepts H^+ and becomes cationic.
Isoelectric pH (pI)	Amino acid exists mainly as zwitterion and has no net charge.
High pH / alkaline medium	Amino acid loses H^+ and becomes anionic.

6. Functions of Amino Acids in the Body

Amino acids are not only the structural units of proteins. They form an active body amino acid pool and are continuously used for growth, repair, metabolism, signaling and nitrogen balance.

Function	Explanation
Protein synthesis	Used to synthesize structural proteins, enzymes, transport proteins, receptors, hormones, immunoproteins and contractile proteins.
Growth and tissue repair	Required for formation and repair of body tissues, especially muscle, connective tissue, enzymes and plasma proteins.
Precursor of nitrogenous compounds	Provide nitrogen and carbon for purines, pyrimidines, porphyrins, heme, creatine, glutathione and other biologically important compounds.
Energy production	During fasting, illness or excess protein intake, amino acids may be oxidized and can contribute about 10-15% of body energy needs.
Glucose and fat formation	Their carbon skeletons may be converted into glucose, glycogen, fatty acids or ketone bodies according to metabolic need.
Neurotransmitters and amines	Glutamate, aspartate and glycine can act as neurotransmitters; decarboxylation forms amines such as histamine, GABA and tyramine.
Detoxification and nitrogen excretion	Amino groups are transferred, deaminated and finally disposed of as urea, helping maintain nitrogen balance.

7. Short Notes: Standard and Non-standard Amino Acids

- Standard amino acids: 20 amino acids repeatedly found in proteins due to the universal genetic code.
- Selenocysteine: Often called the 21st amino acid; found in some selenoproteins such as glutathione peroxidase and thioredoxin reductase.

- Post-translational derivatives: Hydroxyproline and hydroxylysine in collagen; methylated, phosphorylated or acetylated residues in histones.
- Non-protein amino acids: Ornithine and citrulline occur in the urea cycle; GABA functions as an inhibitory neurotransmitter.
- D-amino acids: Present in bacterial cell walls and some antibiotics; they are not the usual building blocks of human proteins.

8. Exam-Oriented Summary

Topic	Must-write point
Definition	Organic compounds with amino and carboxyl groups; protein amino acids are L-alpha-amino acids.
General structure	Alpha-carbon attached to amino group, carboxyl group, H and R group.
Classification	Know structural, polarity, nutritional and metabolic fate classifications.
Essential amino acids	Remember PVT TIM HALL; Arg and His are often treated as semi-essential.
Zwitterion	At isoelectric pH, amino acid carries both positive and negative charge but no net charge.
Main functions	Protein synthesis, tissue repair, precursors, energy, neurotransmitters and nitrogen excretion.